

Teradata

In-Database Graph Functions

Field Developed Solution

Author(s):

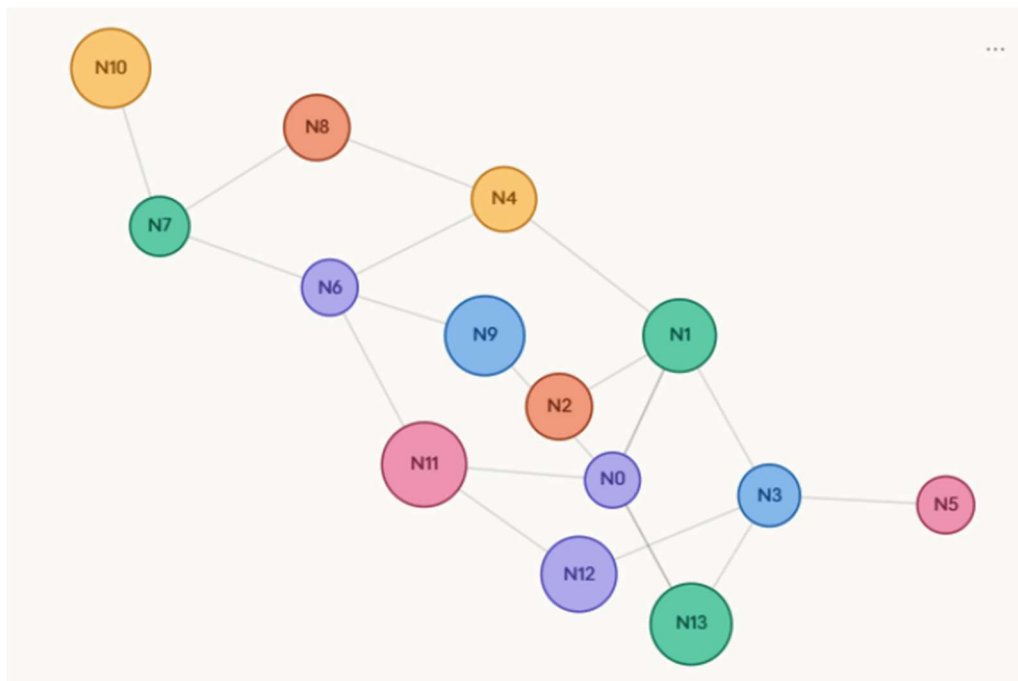
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Executive Summary

EXECUTIVE SUMMARY

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Overview

As organizations manage increasingly interconnected data, including customer relationships, supply chains, fraud networks, and knowledge graphs, the ability to analyze complex relationships at scale has become a strategic priority. Traditional relational databases were not designed for this type of analysis. In-database graph functionality addresses this gap by embedding native graph processing directly within the enterprise data platform and eliminating the need to extract, transform, and move data to a separate graph engine.

The Problem

Relationship-intensive analytics often require organizations to run a separate graph database alongside their primary data platform. This creates friction due to duplicated data, complex pipelines, fragmented governance, and limited ability to join graph results with existing analytical data. The outcome is higher costs, increased latency, and added operational complexity.

The Solution

In-database graph functions bring graph traversal, path analysis, and network analytics directly into the SQL execution environment. Users can express graph queries such as shortest path, community detection, centrality scoring, and subgraph matching using familiar SQL syntax or purpose-built graph extensions, while operating directly on data where it already resides. There is no data movement, no separate system to manage, and no additional cost.

Introduction

Graph analytics unlock insights that traditional relational queries struggle to express, including how entities are connected, how influence or risk propagates through a network, and which nodes hold structural importance across thousands or millions of relationships. For most enterprise data teams, however, graph analysis has required moving data out of the warehouse into a separate specialized system, introducing pipeline complexity, duplicating data, and fragmenting governance.

This solution eliminates that tradeoff.

Teradata Stored Procedures

Each graph algorithm is delivered as a stored procedure that accepts input parameters such as the source table, node and edge column names, a starting node, or a depth limit. The procedure encapsulates all execution logic within Teradata. As a result, graph functions can be invoked like any other database procedure, secured through standard Teradata role management, and versioned and deployed as native database objects.

Python Wrapper

TeradataGE, or Teradata Graph Engine, is a Python package that wraps stored procedure calls and exposes each graph algorithm as a native Python method. It is designed as a companion to the `teradataml` package, which is the official Teradata Python library for in-database analytics, and it follows the same connection model, `DataFrame` conventions, and API style familiar to `teradataml` users.

TeradataGE accepts either a `teradataml` context or an active `TeradataContext` as its entry point, so no separate connection setup is required. Results are returned as `TeradataDataFrame` objects, allowing them to be seamlessly composed with `teradataml` feature engineering, model scoring, and data preparation functions.

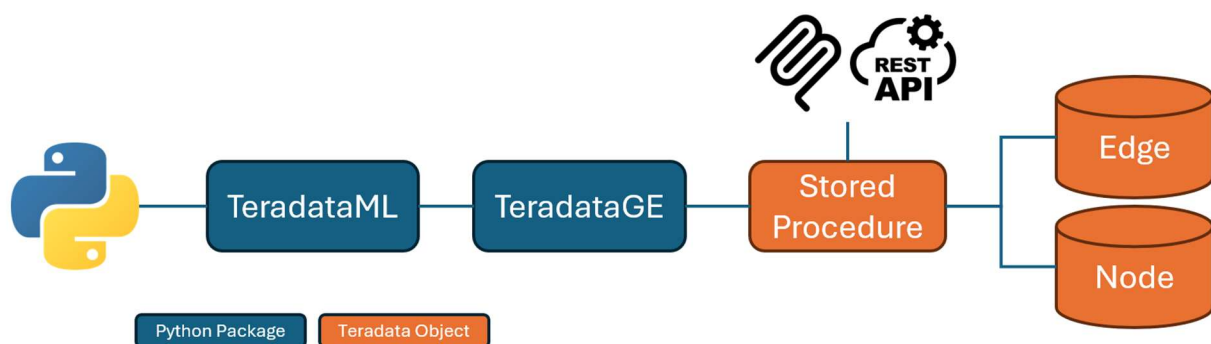


Figure 1: How to use in-database graph functions

Use Cases

Graph analytics is not a single-purpose technology. It is a versatile framework that delivers value across industries and disciplines. The following use cases demonstrate how organizations apply graph analytics to address challenges that traditional relational or tabular approaches struggle to solve.

Anti-Money Laundering (AML)

- Regulatory compliance requires financial institutions to understand not only who their customers are, but also how those customers are connected to others. Graph analytics enables compliance teams to trace complete ownership structures of corporate entities, including shell companies and beneficial ownership chains that may span dozens of jurisdictions.
- In anti-money laundering investigations, graph traversal can follow the flow of funds across multiple hops, exposing layering schemes that are difficult to detect in flat transaction logs. What might take investigators weeks to untangle manually can be revealed in seconds through graph-based path analysis.

Key graph techniques: Graph traversal, hierarchical relationship modeling, centrality analysis.

Network Operations and Optimization

- Telecom networks are, at their core, graph structures, with nodes representing equipment and edges representing connections. Graph analytics allows network engineers to model coverage, simulate failure scenarios, and optimize routing in ways that static topology maps cannot.
- When an outage occurs, graph-based root cause analysis can quickly pinpoint the source and calculate the scope of impact across dependent services. Capacity planning teams also use graph algorithms to identify bottlenecks and evaluate the effects of infrastructure investments before deployment.

Key graph techniques: Shortest path, maximum flow, minimum spanning tree, impact propagation.

Knowledge Graphs and Enterprise Search

- Enterprises accumulate vast amounts of unstructured knowledge across documents, wikis, emails, and databases, which can be difficult to discover and connect. Knowledge graphs organize this information into a structured network of entities and relationships, enabling semantic search and intelligent question answering.
- Organizations are using knowledge graphs to power internal search experiences that deliver not just documents, but contextually relevant answers. When combined with large language models, graph-augmented retrieval, often referred to as GraphRAG, significantly improves the accuracy, grounding, and explainability of AI-generated responses in enterprise environments.

Key graph techniques: Knowledge graph construction, entity linking, graph-augmented retrieval (GraphRAG), ontology modeling.

Entity Resolution

- **Fraud Detection Use Case:** Fraudsters frequently create multiple accounts using slightly altered identities—variations in name spelling, burner phone numbers, shared addresses—and coordinate activity across those accounts. While no single account may trigger a fraud rule in isolation, analyzing the relationships and patterns across the network exposes the fraud ring. Traditional, siloed account-level reviews fail to detect this behavior.
- **Customer Identity Unification Use Case:** A telecom provider maintains customer records across multiple systems—CRM, billing, and customer support—often inherited through acquisitions. As a result, the same real-world customer can appear multiple times under different names, phone numbers, or addresses. Entity Resolution identifies, clusters, and ranks these duplicate entities, creating a unified customer view without relying on supervised labels.

Key graph techniques: PageRank, Community Detection

Social Network Analysis and Influence Mapping

- Whether in marketing, public policy, or investigative research, understanding how information and influence flow through a network is enormously valuable. Graph analytics enables organizations to identify key influencers, detect coordinated inauthentic behavior, and model how ideas or products will spread through a population.
- Marketing teams use influence graph analysis to design more effective seeding strategies for product launches. Public health agencies have applied the same techniques to model disease transmission and optimize intervention strategies.

Key graph techniques: Centrality algorithms (Betweenness, PageRank), Community Detection, and Diffusion Modeling.

Data Lineage and Pipeline Dependency Analysis

- In modern data platforms, raw data passes through dozens of transformation steps, including ingestion jobs, dbt models, Spark pipelines, and BI dashboards, before it becomes a business metric. When a source table changes schema or a pipeline fails, it is essential to understand which downstream reports and models are affected.
- Graph analytics models data assets such as tables, views, models, and dashboards as nodes, with data flow represented as directed edges. This allows data engineers to trace lineage instantly in both directions: upstream to identify the root cause of data quality issues, and downstream to assess the impact of schema changes before deployment.

These capabilities are foundational to reliable DataOps and effective data governance.

Key graph techniques: Directed Graph Traversal, Transitive Closure, Centrality Analysis, Node Classification.

Architecture

Overview

The in-database graph function architecture is built entirely within the Teradata platform. No external graph engine, no data movement, and no additional infrastructure are required. Graph algorithms execute as SQL workloads inside the Teradata engine, leveraging the platform's MPP architecture for parallelism, volatile temporary tables for session-scoped intermediate state, and stored procedures as the callable unit of execution.

The architecture has four layers that work in concert: the **interface layer** (SQL or TeradataGE Python), the **procedure layer** (stored procedures), the **execution layer** (recursive CTEs running across AMPs), and the **storage layer** (source tables and volatile temp tables).

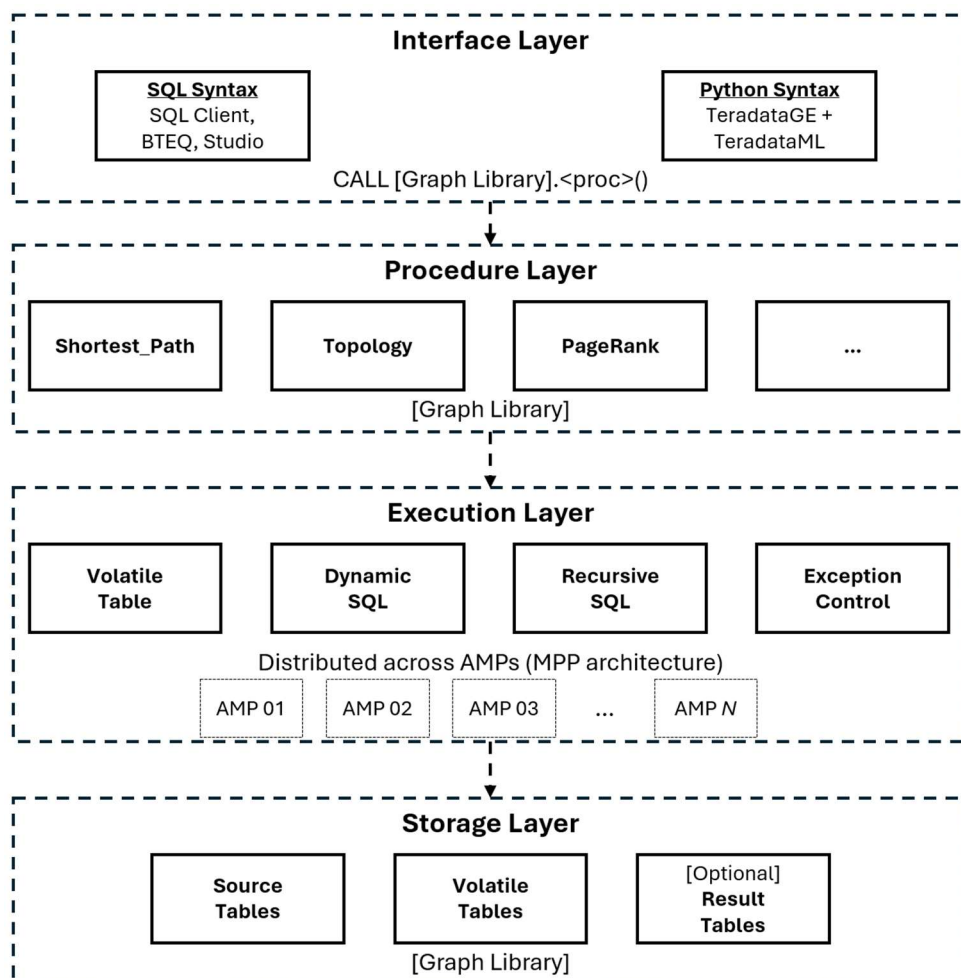


Figure 2: In-database Graph Engine Architecture

Layer 1: Interface Layer

Users interact with the graph library through two equivalent interfaces. Both invoke the same stored procedures and produce identical results.

- **SQL interface:** any SQL client connected to Teradata (Teradata Studio, BTEQ, DBeaver, Jupyter with Teradata SQL kernel) can call graph functions using a standard CALL statement. This makes graph functions accessible to SQL analysts without any Python dependency.
- **TeradataGE Python interface:** the TeradataGE package wraps stored procedure calls as Python methods and integrates with an active teradataml context. Results are returned as TeradataDataFrame objects. This interface is suited for data scientists working in notebooks or automated ML pipelines.
- Both interfaces pass identical parameters to the procedure layer and receive the same output structures.

Layer 2: Procedure Layer

Graph algorithms are deployed as Teradata Stored Procedures in a dedicated database (graph_db). Each stored procedure is responsible for:

- **Validating input parameters:** table names, column names, node identifiers, and algorithm-specific settings such as iteration limits or convergence thresholds
- **Initializing volatile temporary tables:** creating the session-scoped working tables the algorithm requires before execution begins
- **Orchestrating recursive execution:** invoking the recursive CTE logic and managing iteration state across steps
- **Writing final results:** materializing the output into a volatile result table or returning a result set to the caller
- **Cleaning up intermediate state:** dropping working volatile tables at procedure completion to release spool space

Stored procedures are granted execute permissions through standard Teradata role-based access control. A user or role with EXECUTE PROCEDURE privilege on graph library database can call any graph function. No direct table access to intermediate working tables is required or exposed.

Layer 3: Execution Layer

Each stored procedure is constructed with the following components:

Volatile tables

- Each stored procedure creates volatile temporary tables that are scoped to the current database session. These volatile tables exist only for the duration of the session and are automatically dropped when the session ends. The tables are written to Teradata spool space and remain private to the session. As a result, two users running the same graph procedure at the same time each receive an isolated set of working tables with no risk of conflict.

Dynamic queries

- Each SP is building and executing a SQL string at runtime rather than writing it as a fixed statement at compile time.
- It supports dynamic input table structure.

Recursive queries

- Graph traversal is implemented using recursive query. A recursive CTE consists of two parts joined by UNION ALL:
 - **Anchor query:** Defines the starting state of the algorithm (e.g., the seed node at iteration 0, or all nodes initialized with a starting score)
 - **Recursive term:** Defines how the state expands at each subsequent step (e.g., join the current frontier to the edge table to discover next-hop neighbors, or propagate scores to adjacent nodes)

Exception Control

- Exception control in Teradata stored procedures is handled through a structured condition/handler mechanism, similar to try-catch in application languages.
- Some exceptions can be ignored. For example, dropping a table if not exists
- Other critical exceptions will stop the process and return the corresponding SQL error message.

Workload Management via TASM

All graph function stored procedures execute as native Teradata SQL workloads and are fully governed by Teradata Active System Management (TASM). TASM classifies each stored procedure invocation based on configurable criteria, including the calling user, account string, application name, and request characteristics, and then assigns it to the appropriate workload definition. This allows platform administrators to manage the resource consumption of graph analytics workloads using standard TASM controls such as priority tiers, concurrency limits, virtual partition allocation, and response time targets.

Organizations with existing TASM policies can apply the same governance rules to graph workloads without additional configuration. In environments that require intensive or long-running graph processing, such as deep multi-hop path analysis or large-scale PageRank iterations, TASM workload definitions can be tuned to allocate dedicated resources, enforce execution time limits, or queue requests during peak usage periods. This ensures that in-database graph analytics integrates seamlessly into the overall workload management strategy of the Teradata platform.

MPP Parallelism

Teradata distributes table rows across AMPs according to each table's Primary Index (PI). In-database graph functions are designed to align edge table distribution with the traversal access pattern:

- Edge tables are recommended to be PI-distributed on the source node column. This ensures that all outgoing edges for a given node reside on the same AMP, allowing local join during each traversal step and minimizing inter-AMP redistribution.
- At each recursive iteration, the engine expands the frontier in parallel across all AMPs simultaneously. Each AMP processes the subset of nodes and edges it holds, and partial results are merged across AMPs before the next iteration begins.
- For algorithms that require global aggregation, such as PageRank score normalization or convergence checks, Teradata's built-in parallel aggregation performs cross-AMP reduction without the need for application-level coordination.

This design means that graph traversal scales horizontally with the Teradata system. Larger graphs are handled by distributing more data across more AMPs, not by adding external graph infrastructure.

Layer 4: Storage Layer

Source Tables

Graph functions operate on standard Teradata permanent tables. The typical inputs are:

Table	Contents	Optional	Recommended PI
Edge	One row per directed relationship (source node, target node)	N	Source Node ID
	Weight / Probability	Y	
	Edge Type	Y	
	Edge Attribute(s)	Y	
Node	One row per node	Y	Node ID
	Node Name	Y	
	Node Type	Y	
	Node Attribute(s)	Y	

Result Table

- Optional Result Table can be created at the end of Stored Procedure process. If the SP call does not include output table, the result set will be returned directly from SP.
- Result table can be either perm or volatile table.

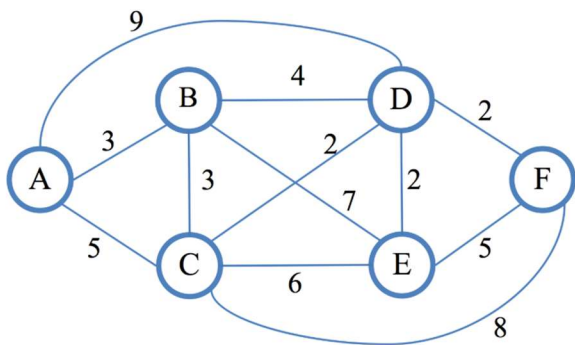
Note:

- The Node ID column in both the Edge table and Node table must be defined as either INTEGER or BIGINT data type. Other numeric or character types are not supported and will result in an error during execution.

Data

In-database graph analysis requires at least one table or view as edge input. For visualization purposes, it is highly recommended to also have a node table or view.

Data Structure:



Integer	VARCHAR		
NodeID	NodeName	Attr1	Attr2
1	A		
2	B		
3	C		
4	D		
5	E		
6	F		

Node Table

Integer	Integer	Float			
FromID	ToID	Weight	Attr1	Attr2	Attr3
1	2	3			
1	3	5			
1	4	9			
2	3	3			
2	4	4			
2	5	7			
3	4	2			
3	5	6			
4	5	2			
4	6	2			
5	6	5			

Edge Table

Implementation

Prerequisites:

- **Database:**
 - Teradata database V17.20 or above.
 - A database *[GraphDB]* for Stored Procedure Installation.
 - Database Privileges for installer.
 1. GRANT CREATE Procedure on *[GraphDB]* to *[Installer]*;
 2. GRANT EXECUTE Procedure on *[GraphDB]* to *[GraphDB]*;
 3. GRANT ALL on *[TargetDB]* to *[GraphDB]* with grant option;
 - Database Privileges for user.
 1. GRANT EXECUTE Procedure on *[GraphDB]* to *[User]*;
- **Client:**
 - Python 3.8 or above
 - The following packages are required:
 1. teradataml>=20.0.0.9
 2. teradatasql>=20.0.0.40
 3. pandas>=2.2.0
 4. pathlib>=1.0.1
 5. plotly>=6.3.1

Installation:

- Download and install TeradataGE python package from PyPI:
 - Command: **pip install teradatage**
- Install Graph Analysis SPs from python with TeradataML and TeradataGE packages:

```
# Teradata python package
from teradataml import *
import getpass
from TeradataGE import td_graph_function, configure, install_db_objects

# Database locator: where graph functions are being installed
configure.graph_install_location = "GraphLib"

# Connect to Teradata database
hostname = "xx.xx.xx.xx"
user = input(prompt="Username for SPs instalation:")
password = getpass.getpass(prompt="Database password for {user}:")
logmech = "TD2"
eng = create_context(host = hostname, username = user, password = password, logmech = logmech)
print(eng)

# Install stored procedures
install_db_objects.install_graph_functions()

# Disconnect from database
remove_context()
```

How to use

Class: Graph Object

TeradataGE's "**td_graph_object**" defines the schema for graph object. Graph object must be defined by calling the corresponding functions. Here are parameters:

Parameter	Description	Default
database_name	Databasename for source and target tables	Current database
edge_table_name	Table/view name for EDGE	
edge_from_node_column_name	Column name for the source node id in edge table	
edge_to_node_column_name	Column name for the target node id in edge table	
edge_type_column_name	Column name for Edge Type	None
edge_weight_column_name	Column name for weight or probability	None
datetime_column_name	Date or Time stamp column for temporal process	None
weight_type	If weight column is defined, weight type should be W (for weight) or P (for probability).	W
edge_attributes	List of attributes column(s) for edge	None
node_table_name	Table/view name for NODE	None
node_id_column_name	node id column name	None
node_type_column_name	node type	None
node_label_column_name	Node name label (for display)	None
source_id	List of starting node id (can be added later)	None
target_id	List of targeting node id (can be added later)	None

Example 1: Weighted Network with Edge only

```
graph_obj = td_graph_function.td_graph_object(edge_table_name = 'graph_edge',
                                              edge_from_node_column_name = 'from_node',
                                              edge_to_node_column_name = 'to_node',
                                              edge_weight_column_name = 'edge_weight'
                                              )
```

Example 2: Unweighted Network with Node and Edge

```
graph_obj = td_graph_function.td_graph_object(edge_table_name = 'imdb_graph_edge',
                                              edge_from_node_column_name = 'from_id',
                                              edge_to_node_column_name = 'to_id',
                                              node_table_name = 'imdb_graph_node',
                                              node_id_column_name = 'node_id',
                                              node_label_column_name = 'primaryname'
                                              )
```

Function: Node and Edge information

Function: edge_info & node_info

Description: Once the graph object is defined, both edge and node information can be retrieved.

Parameter: None

Example:

Python	# Display the defined information for edge print(graph_obj.edge_info())
Return	Edge Table [imdb_graph_edge]: From[from_id] To[to_id] with No Weight. # of edges = 99600 & # of nodes used = 17989.
Python	# Display the defined information for node print(graph_obj.node_info())
Return	Node Table [imdb_graph_node]: Node ID [node_id]. # of Nodes = 17989.

Function: Set Source IDs

Function: set_node_id_from_label_source

Description: Find and set the source ID(s) from node description

Parameter	Description	Default
label	Source ID search based on node's label. '%' can be used for matching sequence. "ALL" represents all the node possible IDs.	
Nodetype	Search based on specific node type.	None
If_exists	replace: replace existing source IDs. append: added into previous existing source IDs.	replace
show_query	Show the actual query	False

Example:

Python	graph_obj.set_node_id_from_label_source("The Last of the Mafia")
Python	graph_obj.set_node_id_from_label_source("%Mafia%")

Function: Set Target ID

Function: set_node_id_from_label_target

Description: Find and set the target ID from node description

Parameter	Description	Default
label	Target ID search based on node's label. '%' can be used for matching sequence.	
show_query	Show the actual query	False

Example:

Python	graph_obj.set_node_id_from_label_target("Midnight Madness")
---------------	---

Function: Topological Order (Traversal)

Function: topology

Description: Find and set the target ID from node description

Parameter	Description	Default
source_id	Starting Node IDs. If not defined, use the existing source IDs	None
edge_pattern	Specific each pattern. If none, use all possible edges	None
max_path_length	It is a traversal depth-limiting parameter used in graph engines to control how many levels (hops) deep a graph traversal algorithm will explore from a starting node.	100
weight_filter	Optional search condition based on weight column. Example: '<0.01'	None
return_data	What kind of data is returned at the end. P : network paths N : Impacted nodes	P
output_table	Name of output table. If none, result will be returned from SP directly	None
temp_output_table	True: Output as volatile temporary table False: Output as perm table	True
show_query	Show the actual query	False

Example:

Python	<pre>graph_obj = td_graph_function.td_graph_object(database_name = 'graphdb', edge_table_name = 'graph_edge', edge_from_node_column_name = 'from_node', edge_to_node_column_name = 'to_node', edge_weight_column_name = 'edge_weight', node_table_name = 'graph_node', node_id_column_name = 'id', node_label_column_name = 'node_name') # Directed Path all_path_from_1node = graph_obj.td_topology(source_id = 1, return_data = 'P', output_table = 'example_topology_path_vt') all_path_from_1node</pre>																																	
Return	<table><tr><th>fullpath</th><th>weight</th><th>path_level</th></tr><tr><td>1,3,5</td><td>5.8</td><td>2</td></tr><tr><td>1,3,5,7</td><td>12.8</td><td>3</td></tr><tr><td>1,2,5,7</td><td>12.8</td><td>3</td></tr><tr><td>1,3,6,7</td><td>9.0</td><td>3</td></tr><tr><td>1,2,4,7</td><td>16.5</td><td>3</td></tr><tr><td>1,2,4</td><td>6.5</td><td>2</td></tr><tr><td>1,3,6</td><td>6.5</td><td>2</td></tr><tr><td>1,2</td><td>3.0</td><td>1</td></tr><tr><td>1,3</td><td>3.0</td><td>1</td></tr><tr><td>1,2,5</td><td>5.8</td><td>2</td></tr></table>	fullpath	weight	path_level	1,3,5	5.8	2	1,3,5,7	12.8	3	1,2,5,7	12.8	3	1,3,6,7	9.0	3	1,2,4,7	16.5	3	1,2,4	6.5	2	1,3,6	6.5	2	1,2	3.0	1	1,3	3.0	1	1,2,5	5.8	2
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Function: Shortest Path

Function: td_shortest_path

Description: Find the shortest path from source node to target node with or without weight

Parameter	Description	Default
source	INT: Source node ID STR: Source node label	None
target	INT: Target node ID STR: Target node label	None
max_path_length	It is a traversal depth-limiting parameter used in graph engines to control how many levels (hops) deep a graph traversal algorithm will explore from a starting node.	100
output_table	Name of output table. If none, result will be returned from SP directly	None
temp_output_table	True: Output as volatile temporary table False: Output as perm table	True
show_query	Show the actual query	False

Example:

Python	<pre>graph_obj = td_graph_function.td_graph_object(database_name = 'graphdb', edge_table_name = 'graph_edge', edge_from_node_column_name = 'from_node', edge_to_node_column_name = 'to_node', edge_weight_column_name = 'edge_weight', node_table_name = 'graph_node', node_id_columnn_name = 'id', node_label_columnn_name = 'node_name') # Shortest Path graph_obj.td_graph_path_decode(source = 1, target = 5, edge_attributes = 'edge_weight', include_plot=True) all_path_from_1node</pre>									
Return	<table><tr><th></th><th>fullpath</th><th>weight</th></tr><tr><td>0</td><td>1,2,5</td><td>5.8</td></tr><tr><td>1</td><td>1,3,5</td><td>5.8</td></tr></table>		fullpath	weight	0	1,2,5	5.8	1	1,3,5	5.8
	fullpath	weight								
0	1,2,5	5.8								
1	1,3,5	5.8								

Function: Path Decode

Function: `td_graph_path_decode`

Description: Convert output path with node label

Parameter	Description	Default
input_table	Path table name, it's created by either <code>td_topology</code> or <code>td_shortest_path</code> call. If the tablename is not specified, the output table from last call will be used	None
input_table_kind	V: Volatile T: Perm Table	V
edge_attributes	Information included between nodes	None
output_table	Name of output table. If none, result will be returned from SP directly	None
temp_output_table	True: Output as volatile temporary table False: Output as perm table	True
include_plot	Include a Sankey diagram as output	False
show_query	Show the actual query	False

Example:

Python	<pre>all_path_from_1node = graph_obj.td_topology(source_id = 1, return_data = 'P', output_table = 'example_topology_path_vt', temp_output_table = True) path_label_df, fig = graph_obj.td_graph_path_decode(edge_attributes = 'edge_weight', include_plot=True) fig</pre>
Return	

Function: PageRank

Function: td_pagerank

Description: IT measures the importance or influence of each vertex (node) in a graph based on the structure of incoming edges

Parameter	Description	Default
directed	True: Directional network False: Bi-directional network	True
damping	Probability of following a link vs. jumping randomly.	0.85
max_iterations	Maximum number of iterations to run	100
tolerance	Stops when score change per iteration falls below this value	1e-8
show_query	Show the actual query	False

Example:

Python	<pre># Directed + Unweighted network graph_obj = td_graph_function.td_graph_object(edge_table_name = "graph_edges", edge_from_node_column_name = "source", edge_to_node_column_name = "target", edge_weight_column_name = None) df_du, iters_du = graph_obj.td_pagerank() print(df_du)</pre>																																	
Return	<table><thead><tr><th></th><th>node</th><th>pr_score</th></tr></thead><tbody><tr><td>23970</td><td>1</td><td>0.000156</td></tr><tr><td>21206</td><td>3</td><td>0.000103</td></tr><tr><td>45503</td><td>4</td><td>0.000089</td></tr><tr><td>48232</td><td>2</td><td>0.000087</td></tr><tr><td>11785</td><td>11</td><td>0.000086</td></tr><tr><td>17638</td><td>7</td><td>0.000083</td></tr><tr><td>8356</td><td>177</td><td>0.000082</td></tr><tr><td>38627</td><td>10</td><td>0.000073</td></tr><tr><td>17568</td><td>26</td><td>0.000068</td></tr><tr><td>19328</td><td>5</td><td>0.000066</td></tr></tbody></table>		node	pr_score	23970	1	0.000156	21206	3	0.000103	45503	4	0.000089	48232	2	0.000087	11785	11	0.000086	17638	7	0.000083	8356	177	0.000082	38627	10	0.000073	17568	26	0.000068	19328	5	0.000066
	node	pr_score																																
23970	1	0.000156																																
21206	3	0.000103																																
45503	4	0.000089																																
48232	2	0.000087																																
11785	11	0.000086																																
17638	7	0.000083																																
8356	177	0.000082																																
38627	10	0.000073																																
17568	26	0.000068																																
19328	5	0.000066																																

Function: Louvain Communities

Function: `td_louvain_communities`

Description: The Louvain method is an unsupervised graph algorithm that automatically finds clusters of nodes that are more densely connected to each other than to the rest of the network.

Parameter	Description	Default
directed	True: Directional network False: Bi-directional network	True
p_threshold	Minimum weight for the relationship	0.5
max_iterations	The maximum number of times Louvain repeats its two phases	100
p_resolution	The resolution parameter controls the granularity of communities. Low resolution (0.5) → fewer, larger communities High resolution (1.5) → more, smaller communities	1.0
output_table	Name of the output table	
temp_output_table	Output as Volatile temporary table	True
show_query	Show the actual query	False

Example:

Python	<pre> inp_tbl = "RLdata10000_select" out_tbl = "RLdata10000_pr_score" # Construct Graph graph_obj = td_graph_function.td_graph_object(edge_table_name = inp_tbl, edge_from_node_column_name = "rec_id_1", edge_to_node_column_name = "rec_id_2", edge_weight_column_name = "score") # PageRank df_du, iters_du = graph_obj.td_pagerank(directed = True, damping = 0.85, output_table = out_tbl, temp_output_table = False) # Louvain Community out_tbl = "RLdata10000_comm" df_comm, community = graph_obj.td_louvain_community(directed = True, p_threshold = 0.7, max_iterations = 100, p_resolution = 1.5, output_table = out_tbl, temp_output_table = False) </pre>
Return	<pre> {'iterations': 100, 'communities': 4697, 'node': 7304, 'modularity': 0.13631219393021848} </pre>

Conclusion

In-database graph functions enable organizations to analyze complex, interconnected data directly within the Teradata engine, eliminating the need for external graph systems and unifying data management with advanced network analytics.

This material covered key in-database graph capabilities, including traversal algorithms, PageRank-based influence analysis, and structural techniques such as path finding, dependency resolution, and pattern matching, all designed to operate efficiently at enterprise scale.

The `teradataml` and `TeradataGE` packages make these capabilities accessible to Python data scientists through a familiar, `DataFrame`-based API, without requiring deep expertise in graph theory or low-level SQL optimization. Additional graph functions can be introduced based on customer requirements and use case demand.

Key Takeaways

01

PERFORMANCE

Algorithms run inside Teradata;
no data movement, no external
graph database needed.

02

SCALABILITY

Handles billions of edges using
Teradata's massively parallel
processing (MPP) architecture.

03

SIMPLICITY

Python-native API via `teradataml`
makes graph analytics
accessible to data scientists.

04

VERSATILITY

One engine covers traversal,
ranking, pathfinding, and
dependency analysis across all
domains.

About Teradata

At Teradata, we believe that people thrive when empowered with trusted information. That's why we built the most complete cloud analytics and data platform for AI.

By delivering harmonized data, Trusted AI, and faster innovation, we uplift and empower our customers—and our customers' customers—to make better, more confident decisions. The world's top companies across every major industry trust Teradata to improve business performance, enrich customer experiences, and fully integrate data across the enterprise. We drive positive impact for hundreds of millions of people every day around the world with faster, flexible data integration and trusted, cost-effective AI innovation.

